

Project ID :

1. Topic (12 words max)

IoT & ML-based Vehicle Speed Monitoring with RFID-Enabled License Validation

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max)
– references not included in word count.

Over-speeding continues to be one of the major causes of highway-related traffic accidents and fatalities in Sri Lanka and globally. According to the World Health Organization, over-speeding contributes to more than 30% of road fatalities worldwide, and the lack of proactive speed enforcement mechanisms exacerbates the issue in developing countries [1]. In Sri Lanka, conventional methods such as handheld speed guns and speed cameras are only partially effective. These systems are reactive, limited in coverage, dependent on manual monitoring, and offer minimal predictive insight into high-risk driver behavior or vehicle categories.

More importantly, authorities lack data-driven tools to identify frequent offenders or predict which roads and times are most susceptible to violations. This gap in proactive monitoring prevents traffic enforcement agencies from deploying resources effectively and hinders the development of policy interventions targeting repeat violators.

To address these challenges, this research proposes the integration of RFID-based tracking using tags embedded into driving licenses, supported by IoT and machine learning technologies. RFID has already been used effectively in applications such as electronic toll collection and vehicle identification in countries like India and Singapore [2], demonstrating its reliability and scalability.

In this project, RFID readers placed at fixed points on highways will scan license-embedded RFID tags, calculating the vehicle's speed based on timestamps. When speed violations occur, real-time alerts will be sent to traffic enforcement authorities. Furthermore, by collecting violation data over time, we aim to train machine learning models to analyze patterns—such as which types of vehicles or license classes are more likely to violate laws. This predictive component enables targeted awareness campaigns and smarter resource allocation.

This research bridges a crucial gap between real-time enforcement and long-term risk forecasting, creating a holistic system for safer highways in Sri Lanka. It combines established RFID technology with predictive analytics to form an innovative and scalable solution to a long-standing traffic enforcement challenge [3].

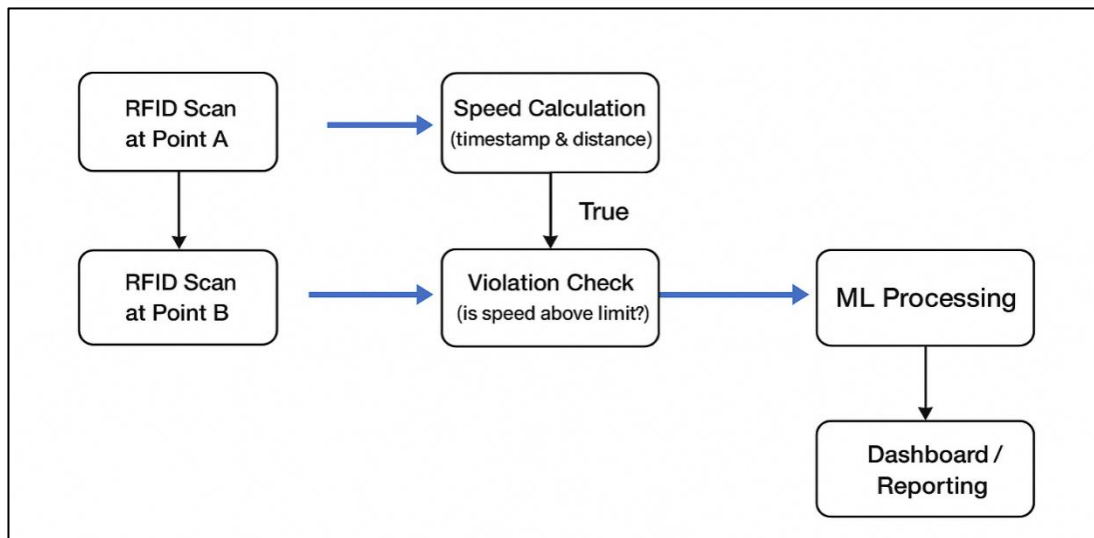
[1] World Health Organization. "Global status report on road safety 2018." World Health Organization, 2018.

[2] FASTag by NHAI, Ministry of Road Transport & Highways, Government of India.
<https://www.nhai.gov.in>

[3] K. Chawla, et al. "An IoT-based Smart Traffic Management System for Smart Cities," IEEE Internet of Things Journal, 2021.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is a smart highway speed enforcement and data-driven traffic monitoring system that combines IoT technology with machine learning for real-time violation detection and predictive analysis. The system assumes that every driver possesses a government-issued RFID-enabled driving license which can be scanned by RFID readers deployed along high-speed highway segments.



As illustrated in the conceptual diagram, the solution begins with two RFID checkpoints placed at a known distance along the road. Each checkpoint reads the RFID tag embedded in the driver's license, capturing a timestamp upon entry and exit. Using the time difference and known distance, the system calculates the vehicle's average speed for that segment.

If the computed speed exceeds the legal limit, the system triggers a violation check, logging the event and sending an automated alert (via SMS or email) to the nearest law enforcement unit. Simultaneously, violation data is forwarded to a machine learning engine that analyzes patterns such as time of day, road segment, vehicle type, and license category.

The ML component builds a predictive model that classifies high-risk drivers and road zones, which are then displayed on an administrative dashboard. This dashboard enables traffic police and transport authorities to visualize violation statistics, generate reports, and make informed decisions for patrol planning and policy enforcement.

By integrating RFID-based vehicle identification, real-time IoT communication, and machine learning intelligence, this solution presents a proactive, scalable, and data-informed solution to highway speed monitoring.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The success of this research project hinges on a multidisciplinary understanding of real-time monitoring technologies, transportation enforcement policies, and intelligent data analytics. A solid grasp of RFID technology, including tag-reader communication protocols, signal ranges, and tag placement best practices, is essential for designing a reliable vehicle identification system. Equally important is the knowledge of highway traffic law and violation categories in Sri Lanka, which guide how speed limits are defined, enforced, and penalized in different road segments.

Additionally, the machine learning component requires expertise in supervised and unsupervised learning techniques. The ability to process structured time-series traffic data and develop classification models capable of identifying frequent violators or high-risk license classes is central to our predictive goal. Selecting the most effective models—such as Random Forest or K-Means Clustering—depends on a strong foundation in statistical analysis and model evaluation.

The backend service development and alert dispatch systems call for proficiency in web technologies, cloud databases, and secure communication protocols. Ensuring fast and reliable data flow from RFID readers to the database, and from the backend to enforcement officials, is crucial for operational success. Moreover, understanding digital security practices is important in safeguarding sensitive driver information.

On the data front, access to realistic traffic flow simulations or test environment logs is necessary. As actual driver license data cannot be used due to privacy concerns, synthetic datasets will be generated during the testing phase to validate system performance. The data must include tag IDs, time logs, speed values, and violation status to effectively train the predictive models. These simulated data structures must mirror real-world diversity in road types, vehicle classes, and driver behaviors to ensure the relevance and scalability of the solution.

8. Objectives and Novelty
Main Objective

To design and develop a smart highway speed monitoring and predictive enforcement system based on the legal assumption that RFID chips are embedded in all government-issued driving licenses. The system will integrate RFID-based driver identification, IoT-based speed tracking, and machine learning techniques to automatically detect speed violations, alert relevant authorities in real time, and analyze behavioral patterns to forecast high-risk driver profiles and road segments for proactive traffic law enforcement.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Lakshan R.H.R IT22917416	To build a machine learning engine that identifies high-risk vehicle types and predicts future speeding behavior based on historical data.	1. Preprocess violation dataset for ML model training 2. Train classification models (e.g., Random Forest) to predict violators. 3. Apply clustering (e.g., K-Means) for driver behavior profiling. 4. Feed predictions into the dashboard via API or JSON	This component provides a data-driven enforcement strategy, enabling authorities to forecast violation trends, prioritize high-risk areas, and personalize enforcement policies using predictive analytics.

Perera A.K.I.R IT22308252	To create a dashboard that visualizes real-time violations, heatmaps, and statistical summaries for traffic authorities.	1. Build secure web interface for admin-level access. 2. Display maps and charts using Google Maps API. 3. Implement filtering (by date, location, ID) and data export (CSV/PDF) 4. Show ML-based violation risk zones on the map	This dashboard bridges enforcement and intelligence by integrating real-time and predictive violation data for enhanced law planning and officer resource management.
Konara K.M.A.K IT22350978	To design the backend service for violation detection, data storage, and real-time alert dispatch to law enforcement.	1. Develop RESTful APIs to receive RFID speed data. 2. Implement speed violation logic per road zone. 3. Store violation records in a secure database (Firestore/MongoDB) 4. Integrate alert system via SMS/email (e.g., Twilio)	The backend enables automated enforcement with no human intervention, creating real-time violation notifications and supporting scalable traffic monitoring infrastructure.

Porogama K K S K IT22156006	To develop an embedded IoT system that calculates vehicle speed using RFID checkpoint scanning of license-tagged drivers.	1. Configure and install UHF RFID readers at two fixed checkpoints along a highway segment. 2. Integrate RFID data with ESP32/Arduino for timestamp logging. 3. Calculate vehicle speed using the difference in entry and exit time. 4. Transmit speed data and tag ID to backend server via HTTP/MQTT	This system introduces a non-intrusive, low-cost highway speed monitoring solution using RFID-based license recognition instead of expensive camera or radar systems, enabling tamper-resistant data capture.
--------------------------------	---	---	---

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Lakshan R.H.R IT22917416	This member handles the machine learning component, including model training, evaluation, and integration into the backend/dashboard pipeline. The work involves data preprocessing, classification (e.g., Random Forest), and clustering (e.g., K-Means). These tasks are fully compliant with the IT specialization's emphasis on data-driven decision-making, predictive analytics, and applied AI.
Perera A.K.I.R IT22308252	This member focuses on building the administrative web dashboard that supports real-time monitoring and reporting. It includes frontend development using modern frameworks (React/Vue), map-based data visualization, filtering mechanisms, and exporting capabilities. This aligns with IT specialization by incorporating UI/UX design, web technologies, and interactive visualization frameworks.
Konara K.M.A.K IT22350978	This member leads the implementation of the server-side logic for violation detection and alert automation. The tasks involve API development, cloud database integration, and secure notification mechanisms (via Twilio/SendGrid). These skills reflect Information Technology specialization areas including backend services, cloud systems, and system interoperability.
Porogama K K S K IT22156006	This member is responsible for the design and development of the RFID-based vehicle speed tracking system using embedded IoT components. The task involves configuring RFID readers, integrating microcontrollers (e.g., ESP32), and transmitting real-time data over the internet. These activities are directly aligned with the core competencies of Information Technology such as embedded systems integration, IoT communication protocols (HTTP/MQTT), and real-time data capture.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.